



OBJECT ORIENTED SOFTWARE ENGINEERING

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4. Evolutionary Process Model:- Prototyping approach

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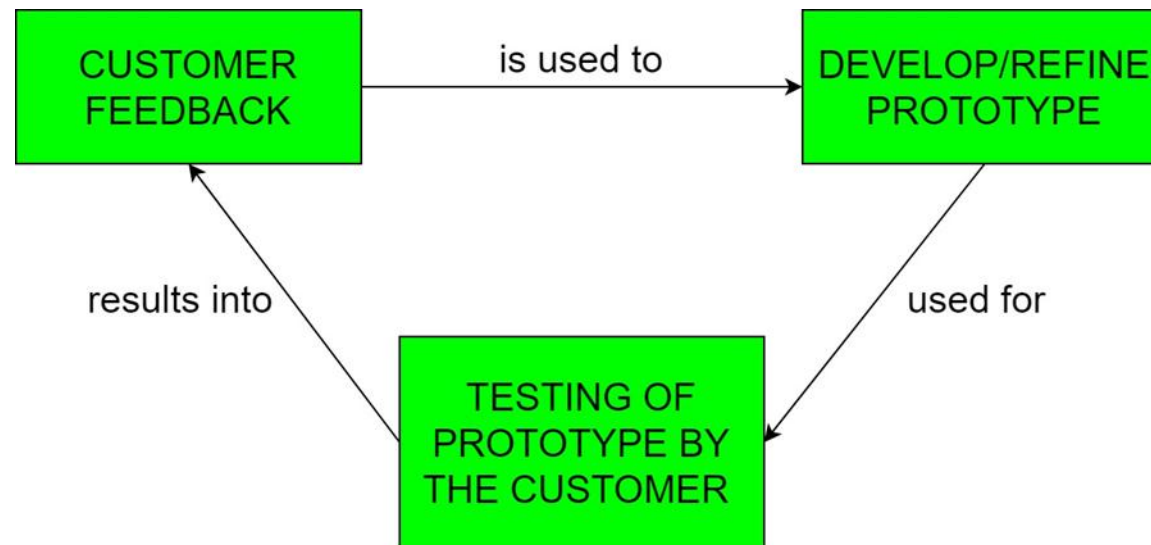
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- Evolutionary Models are **iterative**.
- **Produce** an increasingly more **complete version of the software** with each **iteration**.
- Here the product **evolves over time**.
- Evolutionary models are:
 - **(i) Prototyping**
 - **(ii) Spiral Model**

(i) Prototyping

- Prototyping is defined as the process of developing a **working replication** of a **product** or **system** that has to be **engineered**.
- It offers a small scale facsimile of the end product and is used for obtaining **customer feedback**.

- That is, in this model, a **prototype** of the **end product** is first **developed**, **tested** and **refined** as per customer feedback **repeatedly** till a final acceptable prototype is achieved which forms the basis for developing the final product.

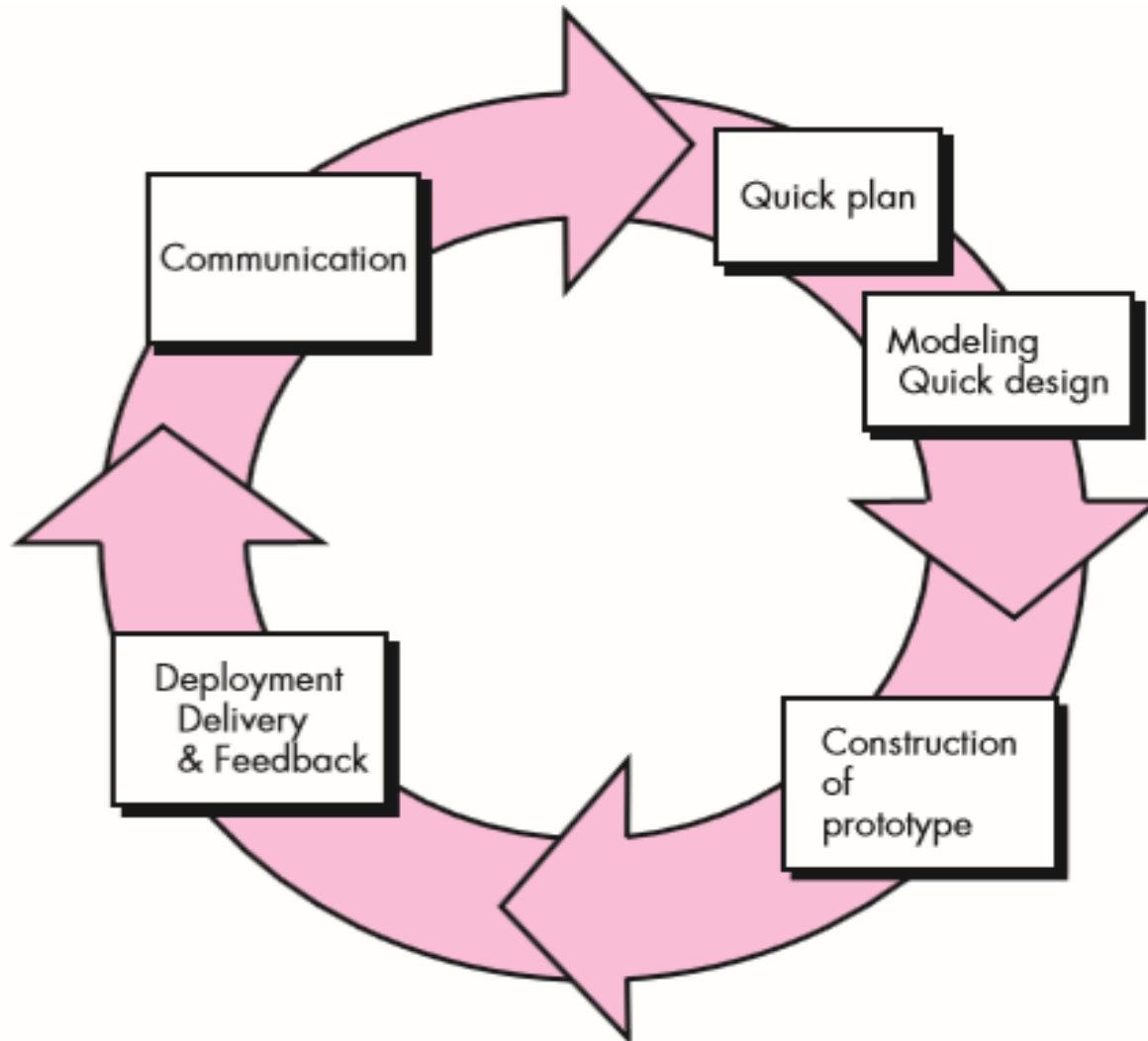


- Prototyping **start with communication**, between a customer and software engineer to define **overall objective, identify requirements and make a boundary**.
- Moving ahead, **quick** planning and **modeling** occurs
- **Quick design** leads to **prototype construction**.

- **Prototype is deployed and evaluated** by the customer/user.
- **Feedback from customer/end user will refine requirements** and that is how **iteration** occurs during prototype to satisfy the needs of the customer.

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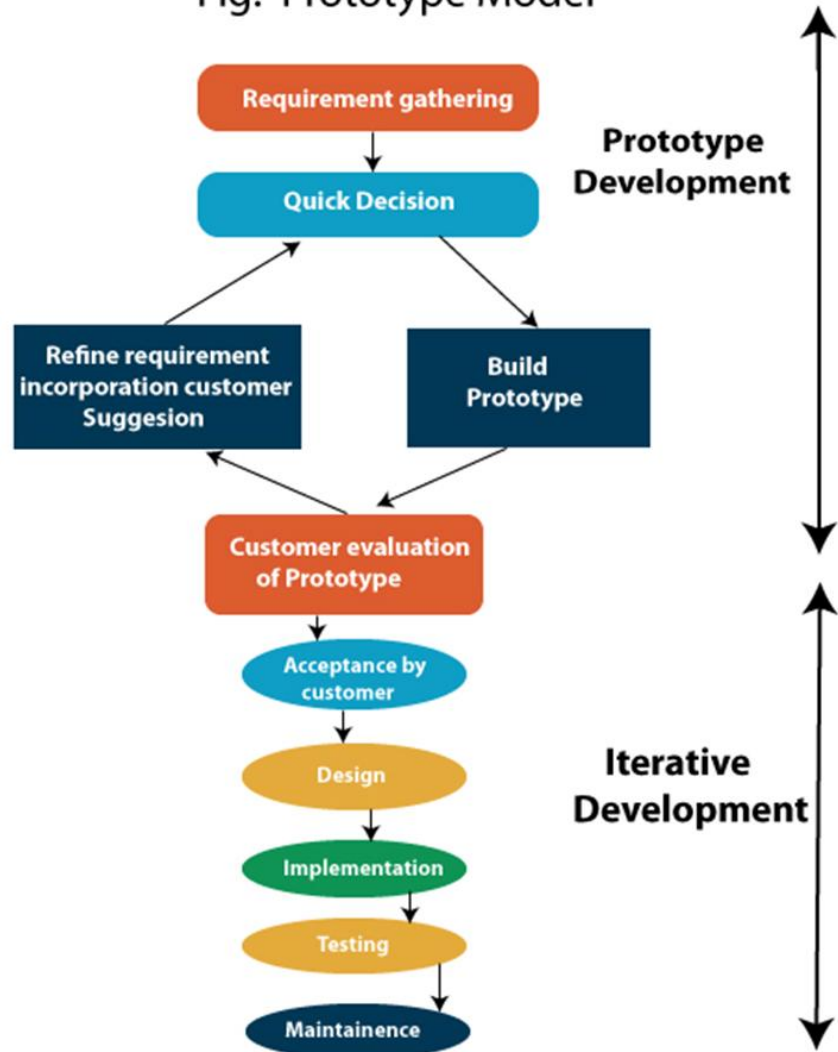
4. Evolutionary Process Model-The Prototyping Paradigm



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4. Evolutionary Process Model-The Prototype Model

Fig: Prototype Model



Advantages:

- Reduced time and costs
- Improved and increased user involvement

Disadvantages

- Insufficient analysis
- User's confusion of prototype and finished system
- Developer's misunderstanding of user objectives
- Developer's attachment towards prototype
- Excessive prototype development time.
- Expense of implementing prototyping



THANK YOU

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